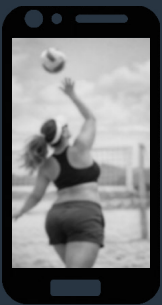
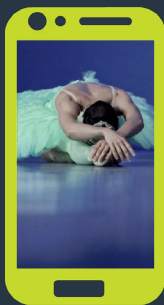
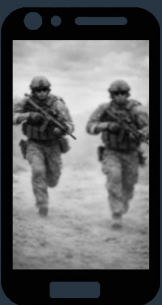
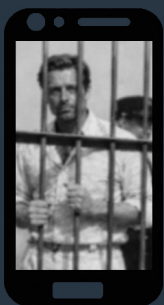
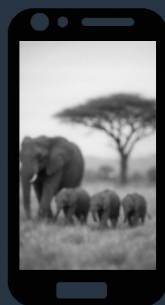
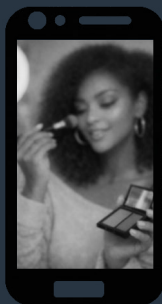
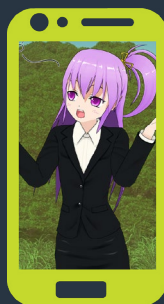
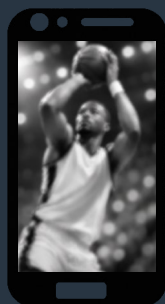


MOBILE PERSONALIZATION

The history and theory of
mobile streaming media



JUSTIN GRANDINETTI
CHARLES ECENBARGER

MOBILE PERSONALIZATION

*The History and Theory of
Mobile Streaming Media*

by
Justin Grandinetti &
Charles Ecenbarger

Mobile Personalization: The History and Theory of Mobile Streaming Media

by Justin Grandinetti & Charles Ecenbarger

© 2026 Justin Grandinetti & Charles Ecenbarger

Works not in the public domain are licensed under a Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). This license allows you to share, copy, distribute and transmit the text; to adapt the text for non-commercial purposes of the text providing attribution is made to the authors (but not in any way that suggests that they endorse you or your use of the work).



Published by:

mediastudies.press

414 W. Broad St.

Bethlehem, PA 18018, USA

Copy-editing: Emily Alexander

Cover design: Natascha Chtena

Landing page: mediastudies.press/mobile-personalization

Media Manifold series - issn (*online*) 2832-6202 | issn (*print*) 2832-6199

isbn 978-1-951399-54-2 (*print*) | isbn 978-1-951399-51-1 (*pdf*)

isbn 978-1-951399-53-5 (*epub*) | isbn 978-1-951399-52-8 (*html*)

doi 10.64629/pf8605cb.10dcf8f3 | lcn 2025944820

Edition 1 published in July 2026

Contents

<i>Introduction: Mobile Matters</i>	<i>1</i>
<i>Chapter One: The History of Mobile Personalization.</i>	<i>9</i>
<i>Chapter Two: Architectures of Mobility</i>	<i>41</i>
<i>Chapter Three: Monetizing Mobility</i>	<i>70</i>
<i>Chapter Four: The Characteristics of Mobile Streaming</i>	<i>93</i>
<i>Chapter Five: The Future of Mobile Streaming</i>	<i>121</i>
<i>Bibliography</i>	<i>128</i>

CHAPTER THREE

Monetizing Mobility

Walk into any coffee shop, airport lounge, or public park, and chances are your phone already knows where you are. And so do the companies behind your apps. Whether it's a push notification for a nearby fast food deal, a Spotify ad tailored to your current zip code, or a TikTok video that just happens to feature a local landmark, mobile streaming has become entangled with location-aware technologies like geofencing. These tools don't just deliver content; they deliver it strategically, turning your physical presence into a point of monetization. This chapter explores the convergence of mobile streaming and forms of monetization, situating the development within broader shifts in how media is produced, delivered, and capitalized on. From a teenager in Lagos scrolling TikTok, to a commuter in Mumbai listening to Spotify, to a Zoom meeting on a park bench in New York, billions of people engage with mobile media as a matter of routine. Platforms like TikTok, Twitch, Instagram, YouTube, and Apple Music function as global infrastructures, systems built not only to entertain and connect but also to shape behavior, collect data, and generate profit, in turn, consolidating power of media engagement. In short, this chapter is about how mobile streaming makes money. This occurs not just through ads, but also through architectures that increasingly know where you are, what you like, and when you're most likely to tap, swipe, or scroll. Such context-aware targeting exemplifies mobile personalization, the tailoring of content and ads to an individual user's data and situation, a concept central to this chapter's analysis.

Last chapter, we detailed how the model of the platform provides insight into how companies like ByteDance (TikTok), Google (YouTube), Meta (Instagram), Amazon (Twitch), and Apple (Apple Music) function as infrastructures of streaming built on data extraction, network effects, and an ecosystem of

control.¹ These services span social media, video, live streaming, and music and are unified by a common mobile personalization logic that turns user engagement into profit. Moreover, the network society situates mobile streaming in a broader transformation where digital networks constitute the social and economic infrastructure of society.² From this perspective, mobile streaming platforms are not just neutral technologies for content delivery; they are highly strategic entities that marshal technical infrastructure, user engagement, and data flows to accumulate capital and influence. This analysis highlights how features like cloud-based delivery, design, cross-platform integration, and the use of application programming interfaces (APIs) all serve to expand the reach and profitability of these platforms. Simultaneously, these same features reshape user experiences and behaviors, making continuous connectivity and consumption a normalized aspect of life.

If the last chapter was all about *how* mobile streaming operates, then this chapter focuses on the *why* as it relates to monetization and the role of mobile personalization in generating profit. Platform and infrastructure providers don't create all this labor and resource intensive architecture out of some form of social good. These companies, of course, operate toward a payoff. But how they do so varies widely. This chapter, consequently, follows the money of streaming and mobile streaming to unearth just how these platforms make a buck or two . . . or a couple billion.

By critically analyzing the monetization strategies of these mobile streaming platforms, we reveal a landscape of platformed media: a complex ecosystem where user attention, personal data, cultural content, and technological infrastructure are all interwoven into new commodity forms. First, we examine how mobile streaming became ubiquitous in everyday life. Through habitual engagement seemingly anytime and anywhere, users funnel their attention, time, and data into the silos of platforms. Next, we delve into five key monetization strategies: (1) mobile and locative aware advertising, (2) mobile subscription and premium services, (3) microtransactions and the creator economy, (4) monetized cross subsidies, and (5) APIs and platform capture. Through this structured analysis, we aim to provide a comprehensive understanding of how contemporary mobile streaming is not merely a set of popular apps, but rather a politically and economically significant development reconfiguring media industries and daily life on a global scale.

¹ Nick Srnicek, *Platform Capitalism* (Polity Press, 2016).

² Christian Katzenbach and Stefan Larsson, "Imagining the Digital Society: Metaphors from the Past and Present," *Digital Society Blog*, Alexander von Humboldt Institute for Internet and Society, May 15, 2017.

Mobile Streaming in Everyday Life: Ubiquity and Habitual Engagement

Before examining how platforms monetize, it's important to understand how deeply mobile streaming is woven into everyday life. The integration of mobile streaming into daily life has reached a point where, for many people, it is taken for granted that any moment of downtime or any routine activity can be augmented with streaming media. Global usage statistics underline just how deeply these platforms have penetrated our routines. For instance, as of 2024, TikTok users worldwide spent an average of ninety-five minutes per day on the app—far more than users spent on any other social network.

³ This is not limited to any single region; TikTok's over one billion monthly active users are spread across Asia, North America, Europe, and beyond. YouTube has over 2.7 billion users and reports that, collectively, people watch over one billion hours of video on YouTube every day.⁴ These figures convey how streaming platforms capture a large share of human attention on a daily basis. Furthermore, Instagram and Facebook together command billions of user visits daily, and services like WhatsApp or WeChat facilitate countless video and voice calls each day around the globe. Even live streaming-centric platforms like Twitch, while more niche, sustain around 35 million daily active users and at any given moment about 2.5 million people are watching a Twitch stream.⁵ Twitch viewers collectively watched over 20 billion hours of content in 2024,⁶ indicating that for a segment of the population, viewing live streams has become a regular part of leisure time.

These numbers translate into lived experiences that are remarkably common across different societies. It is not hard to imagine a morning train ride filled with commuters watching YouTube clips or scrolling Instagram Reels; teenagers coming home from school and broadcasting themselves live on Twitch or TikTok Live; families separated by distance having nightly video chats via WhatsApp or FaceTime; music streaming forming the soundtrack of daily exercise, study, or relaxation. Mobile streaming is embedded in daily rhythms. Many people start their day by checking social media videos or news clips on their phones and end their day watching streaming videos in

³ "TikTok Statistics You Need to Know in 2025," Backlinko, updated January 2026.

⁴ "YouTube Statistics 2025: Users by Country + Demographics," *Global Media Insight*, January 2025.

⁵ *TwitchTracker*, accessed May 12, 2025, <https://twitchtracker.com/>.

⁶ Dhanika Pineda, "Viewers Watched 20.35 Billion Hours on Twitch in 2024: 'Stream Big' Explores Why," *Central Florida Public Media*, December 18, 2024.

bed. In the interim, streaming fills all sorts of micro-moments—waiting in line, eating lunch alone, or taking a work break—as well as more extended periods like evening entertainment. Indeed, streaming media often accompanies activities that used to be free of media or, at least, dictated by broadcast schedules. People now listen to podcasts or music streams while driving or doing house chores, watch recipe videos while cooking, and scroll short videos while riding an elevator. The smartphone has become a constant companion, and the proliferation of high-speed mobile internet means entertainment and communication streams are constantly within arm's reach.

This deep embedding of media into everyday life is both a result of platform strategies and a driver of further growth. Platforms deliberately design their services to foster habitual use. Features like the infinite scroll, such as TikTok's never-ending personalized feed or YouTube's autoplay that queues another video after one finishes, encourage users to lose track of time and consume content continuously. In effect, these are major instances of mobile personalization keeping users hooked. Meanwhile, push notifications draw users back in. YouTube alerts about a channel's new upload, Twitch about a streamer going live, Instagram about a friend's live story, and so on. These notifications are carefully calibrated by algorithms to maximize re-engagement. In other words, the platform learns when and how to prod a user to open the app again. The result is what some commentators call the attention economy at work. That is to say, an environment where various services compete intensely for the finite hours of users' attention because that attention can be monetized.⁷ Social media and streaming platforms have become dominant players in this attention economy by captivating audiences with endless feeds of user-generated content. The more habitual and reflexive the usage becomes ("I'm bored, let me check TikTok" or "It's 8 p.m., time to browse Twitch"), the more stable the platform's audience commodity is for advertisers and the more opportunities for monetization via subscriptions or purchases.

The scale of this phenomenon is a testament to how mobile streaming aligns with certain universal aspects of human behavior, while also being tailored to local contexts. Smartphone adoption has been rapid in both developed and developing countries, which means billions now have a personal screen. In many emerging markets, cheap Android phones and affordable data plans led to a mobile-first internet culture—people whose

⁷ Richard A. Lanham, *The Economics of Attention: Style and Substance in the Age of Information* (University of Chicago Press, 2006).

first and primary way of accessing the internet is through mobile apps. For these users, streaming video on YouTube or local platforms, like India's Hotstar or China's Douyin, effectively replaced legacy media like TV. It goes without saying that large media companies have recognized these opportunities. YouTube, for instance, provides offline download features in markets with patchy connectivity, so users can download videos while on Wi-Fi and watch them on the go, and TikTok launched "TikTok Lite" for lower-end phones. By adapting to local bandwidth and device conditions, platforms further ensured their integration into daily habits across different socio-economic settings.

Moreover, streaming platforms often become integrated with social practices and cultural norms. TikTok's short-video format has given rise to global dance trends, memes, and challenges that engage youth culture; participating in or at least watching these trends becomes a social activity. Twitch and YouTube have created new forms of communal experience. For example, tuning into a live stream of a video game tournament or an influencer's Q&A feels akin to communal TV watching or even attending an event, except it can be done from anywhere on a phone. During the COVID-19 pandemic, usage of video calls and streaming skyrocketed, embedding these platforms even more deeply into work and family life. Zoom, practically a niche business tool pre-2020, became a household name and by April 2020 was hosting 300 million daily participants around the world as people used it for work meetings, remote classes, and virtual social gatherings.⁸ This period underscored how essential streaming communication had become. And even after the pandemic's peak, habits like working remotely via video and maintaining long-distance relationships with video calls have persisted.

The blurring of work and leisure through the same devices and networks is another notable aspect. A smartphone that receives office emails is also the device for watching TikTok; a laptop used for a Zoom meeting at noon might be used to stream Netflix at 9 p.m. This convergence can lengthen the time one stays "on" digital networks, giving platforms even more opportunity to capture user engagement in the margins of daily life (for instance, checking a social app in between tasks while working from home). From the user's perspective, the omnipresence of streaming can be double-edged. While it offers convenience and entertainment on demand, it can also lead to a sense of information overload, or an addictive pull to keep looking at

⁸ "A New Era of Growth: How the Zoom Partner Ecosystem Is Evolving," *Zoom Blog*, Zoom, October 2024.

the screen. Some studies have begun framing compulsive TikTok or social media use in terms of addiction, noting that the algorithmic feed is designed to be hard to put down.⁹

Such design is no accident. It stems from commercial imperatives. Keeping users engaged translates directly into profit, either via more ad impressions or higher likelihood of paying for premium features. This raises questions about the consequences of building daily routines around platforms whose primary goal is profit maximization. Critics argue that when engagement-at-all-cost is the business goal, platforms may encourage unhealthy usage patterns or promote sensational content that grabs attention, such as extreme challenges on TikTok, or outrage-inducing articles, Tweets, or videos, because these drive metrics up, a phenomenon documented across social media algorithms.

Despite these concerns, mobile streaming's integration into everyday life has a self-reinforcing quality. The longer people use these platforms and incorporate them into mundane practices, the more indispensable they feel. For example, a student might rely on YouTube tutorials daily for help with coursework, making YouTube an educational crutch; a professional might habitually listen to industry podcasts on Apple Podcasts or Spotify each morning; a family might develop a routine of a weekly video call across continents. In each case, what starts as a convenience can quickly become seen as a necessity or an assumed background to life. This cultural entrenchment makes it challenging for individuals to opt out of platforms, which, mind you, contributes to their monopoly power, and also creates lock-in on a societal level. In other words, public discourse, trends, and even news increasingly flow through these channels.

The patterns of use encouraged by the platforms—frequent, habitual, and extended engagement—are a direct outcome of their design and business objectives. As we turn next to examine the monetization strategies of these platforms, it is important to keep in mind that the casualness with which we treat ubiquitous streaming is in fact cultivated by design. The omnipresence of streaming is the fertile ground from which platforms harvest revenue. They have made themselves indispensable to users in order to more effectively capitalize on user activity. In short, the habitual engagement cultivated via mobile personalization sets the stage for monetization. Next, we turn to how platforms capitalize on this engagement through five monetization strate-

⁹ Li Gu et al., "What Drives Me to Use TikTok: A Latent Profile Analysis of Users' Motives," *Frontiers in Psychology* 13 (2022).

gies. We break down how exactly these platforms monetize the extensive engagement they have secured, and what new forms of commodification and profit are emerging in the mobile streaming era. This position in daily life underscores the power these platforms wield as gatekeepers of information and entertainment, a power base they will leverage in the monetization strategies that follow.

Monetization Strategies of Mobile Streaming Platforms

Behind the seamless streams of content that characterize the user experience lies a sophisticated array of monetization strategies. Mobile streaming platforms employ multiple business models to turn user engagement into revenue. These strategies include advertising, subscriptions, data monetization and sale, microtransactions and virtual gifting, content partnerships, and/or ecosystem plays. We view each of these strategies as a way to commodify different elements of the streaming experience, from user attention and user data to content and social interactions. This section analyzes the major monetization approaches, highlighting how they work on platforms like TikTok, Twitch, Instagram, YouTube, Spotify, and others.

Mobile and Locative Aware Advertising

Advertising remains one of the most dominant revenue models for mobile streaming platforms, especially those offering free content such as TikTok, Instagram, YouTube, and Twitch (for non-subscribed viewers, though the ad-free experience has slowly been deteriorating even for subscribed viewers). The logic of advertising-based platforms is a direct extension of mass media models from television and radio: Build a large audience and sell access to that audience's attention to advertisers. However, digital streaming platforms have turbocharged this model by leveraging personal data to target ads with precision and by reaching audiences on a one-to-one basis through their personal devices, providing a clear example of mobile personalization being used to maximize effectiveness. The result is a highly profitable system that, as noted earlier, treats user attention as a commodity to be sold. Notably, this personalized ad model spans across platforms, illustrating how a common monetization logic underpins diverse services rather than a single streaming industry. Dallas Smythe's critical insight on the subject is particularly apt: "The audience is the commodity that is bought

and sold and exploited to produce surplus value for advertisers and media companies.”¹⁰ In the mobile streaming context, the audience commodity is fragmented and individualized. Each user is packaged with a profile of data about their likes, habits, and demographics, and advertisers purchase impressions (or influencer partnerships, etc.) aimed at those specific users. In packaging and selling audience data, platforms exercise considerable power over cultural consumption, effectively deciding which voices or trends get visibility based on what’s profitable.

YouTube stands as a prime example. The platform generates tens of billions of dollars in advertising revenue annually by running ads before, during, or alongside videos that billions of people watch daily. Google’s advertising machinery ensures that the ads are tailored. By this we mean that someone who has been watching a lot of cooking videos might see ads for kitchen gadgets, while a heavy consumer of gaming videos might see ads for new video games. Google consistently derives the majority of its income from advertising.¹¹ This heavy dependence makes YouTube very much an advertising platform, meaning its primary business is extracting data and selling ad space. To keep this engine running, YouTube’s algorithms are tuned to maximize watch time and clicks, which in turn maximizes ad exposure. This has raised well-documented concerns about sensational or extremist content being algorithmically promoted—not out of ideological intent, but as a side effect of chasing engagement, which yields more ad impressions.¹² As such, the advertising model can skew what content gets produced and recommended, since creators often chase algorithmic trends to earn advertising dollars or to maintain visibility if the algorithm favors certain content that retains viewers.

Mobile tracking and location awareness are increasingly crucial components to the targeted ad machine. In 2017 US marketers spent more than \$16 billion on targeted mobile advertisements, with a projected growth to \$38.7 billion by 2022.¹³ Location-targeted advertisements have existed in various forms for years, as Facebook offered location-based advertisements for business beginning in 2014. This location-based targeting works through

¹⁰ Dallas W. Smythe, “Communications: Blindspot of Western Marxism,” *Canadian Journal of Political and Social Theory* 1, no. 3 (1977).

¹¹ “Earnings 2025,” Investor Relations, Alphabet Inc., accessed May 12, 2025.

¹² Alex Russell, “YouTube Video Recommendations Lead to More Extremist Content for Right-Leaning Users, Researchers Suggest,” *In Focus* (blog), UC Davis, December 13, 2023.

¹³ Janelle Nanos, “How Companies Use Geolocation Data to Target You,” *The Boston Globe*, 2017.

location sharing services that are built into certain phone features. Businesses provide advertising companies like Meta and Google with their address and preferred geographic radius, and these platforms, in turn, provide businesses with the ability to refine the search to target certain demographics.¹⁴ Though not exclusive to mobile streaming, forms of located-based advertisement are certainly more relevant to mobile devices than fixed, in that advertisements are able to “follow” the user through space. To this end, targeted streaming advertisement offers yet another critical evolution for mobile streaming practices. Put another way, platforms have utilized experiments in mobile streaming to capture user attention through emphasis on sociability, but the goal of these developments resides in strategies of monetization.

Critical to mobile models of advertisement-based video-on-demand is the collection of location, device, and geographic data. Global positioning data can pinpoint user locations, but location-based advertisement accuracy is often limited. However, not all location-aware applications function the same way. Some applications (such as weather apps) only take a snapshot of user location data, while others (including most streaming apps) require more frequently updating user data. For many streaming providers, this geolocative data is necessary to determine whether content can be viewed in that area; streaming platforms like Netflix have different content licenses in various countries, while streaming sports providers often must comply with local regulations and blackouts.

For social media streaming, location data is mainly used for advertising purposes. Potentially troubling is that users are often unable to opt out of disclosing their location, as is the case with Facebook.¹⁵ Early studies of GPS-enabled location-based advertisement noted that most users found the advertisements useful and were unconcerned or unaware of potential privacy issues accompanying location-data collection.¹⁶ Studies show that users believe targeted ads based on fine-grained data collection is incredibly invasive.¹⁷ A 2019 Pew Research Center study similarly found that 81% of US adults believed they had little control over their data, and 79%

¹⁴ Zach Miners, “Facebook’s Location-Based Ads Pitch You on the Store You Just Walked By,” *PC World*, October 7, 2014.

¹⁵ Alex Hern, “Facebook Users Cannot Avoid Location-Based Ads, Investigation Finds,” *Guardian*, December 19, 2018.

¹⁶ Eija Kaasinen, “User Needs for Location-Aware Mobile Services,” *Personal and Ubiquitous Computing* 7, no. 1 (2003).

¹⁷ Nicolette Conti et al., “When Did My Mobile Turn into a ‘Sellphone’?: A Study of Consumer Responses to Tailored Smartphone Ads,” in *Proceedings of the 26th Annual BCS Interaction Specialist Group Conference on People and Computers* (BCS, 2012).

were concerned about how companies were using data collected about their activities, with 72% reporting that they feel that almost all their mobile internet activity was being tracked.¹⁸ Taken together, these findings echo the claim that, for users, when context-aware advertising is made more tangible via smartphones and mobile devices, it is often perceived as an breach of personal space.¹⁹ Consequently, location-aware advertising—including new mobile streaming forms—are often perceived as an invasive threat to privacy for users.

Mobile-oriented platforms like TikTok and Instagram Reels similarly rely on advertising, particularly as they have rapidly grown user bases. TikTok's monetization initially lagged its user growth, but by the mid-2020s it had developed a robust advertising platform. Brands can pay for in-feed video ads that appear as one scrolls, which are often made to look like regular TikToks, or for hashtag challenges and other sponsored trends. The appeal for advertisers is the extremely high engagement and the algorithm's ability to pinpoint niche communities. TikTok collects detailed data on what each user watches, likes, or skips, enabling fine-grained targeting. While much of TikTok's algorithm is proprietary and has not been released, users have deduced that captions, lighting, and other details within the video itself also play a role in the algorithm. In effect, TikTok sells access to moments of user attention when they are most immersed. For instance, an ad might be inserted after a particularly engaging video when the user is likely to continue swiping. Instagram, integrated with Facebook's ad system, offers targeted advertising in Reels and Stories, benefiting from Meta's vast trove of user data across Facebook/Instagram/WhatsApp. Instagram's ad business is a cornerstone of Meta's revenues.²⁰

Advertising in live streaming (Twitch) and music streaming also deserves mention. Twitch inserts video or banner ads on streams, though it has a somewhat more complex relationship with ads because excessive ads can drive viewers away from a live experience, in that missing moments in a live broadcast due to an ad is frustrating for many viewers. Twitch has tried to balance this with features like picture-in-picture ads that run a small ad while keeping the stream visible. Nonetheless, advertising is a significant

¹⁸ Brooke Auxier et al., "Americans and Privacy: Concerned, Confused and Feeling Lack of Control over Their Personal Information," Pew Research Center, November 15, 2019.

¹⁹ Eric Gordon and Adriana de Souza e Silva, *Net Locality: Why Location Matters in a Networked World* (Wiley-Blackwell, 2011).

²⁰ Akash Sriram, "Instagram to Make Up More Than Half of Meta's US Ad Revenue in 2025, Report Shows," *Reuters*, December 18, 2024.

revenue component for Twitch, and it has recently pushed streamers to run more ad breaks by offering higher revenue share incentives.²¹ For a 2019 event called Twitch Sells Out: A Prime Day Special Event, the Amazon-owned streaming platform recruited streaming celebrities for a two-day live shopping showcase for products on sale during Amazon's Prime Day. Twitch derives a substantial percentage of views from mobile users. It is unsurprising, then, that Amazon would utilize Twitch for advertising and cross-promotion, as surveys have found that mobile gamers are particularly receptive to advertising about products they might want.²² Embedded mobile ads have become so prevalent that they have altered the length of content itself. A few years ago, YouTube videos began to increase average run time to facilitate more ads per video. This occurred despite user preferences for shorter content. Put another way, the inbuilt goal of advertising and monetization to YouTube's algorithm was, at least for a while, powerful enough to override human preferences for shorter content and persuade streamers to produce longer-form videos, even in an increasingly mobile environment.

On music platforms, advertising is typically used in a freemium model: For example, Spotify's free tier plays ads periodically, nudging users to pay for premium to remove ads. Apple Music notably eschews ads entirely in favor of a pure subscription model. Apple uses the absence of ads as a selling point and, likely, also as a way to differentiate itself as a more privacy-conscious service, aligned with Apple's device sales strategy. The absence or presence of advertising can shape user experience significantly. Ad-based services bombard users with commercial messages, whereas subscription ones promise uninterrupted content for a fee. As we'll see, many platforms now mix both models.

Mobile Subscription and Premium Services

The second major monetization strategy is subscriptions. Subscription-based monetization offers an alternative or complementary approach to advertising. Under a subscription model, users pay a recurring fee for access to content, often with added perks like no ads or exclusive material. This model has deep roots in older media, such as cable TV packages, premium channels, subscription newspapers, etc., but has been revitalized by streaming, partly

²¹ Taylor Hatmaker, "Twitch Plans to Cut Subscription Revenue for Some Top Streamers in Push Toward Ads," *TechCrunch*, September 21, 2022.

²² Robert Williams, "Twitch Readies 2-day Live Shopping Show for Amazon Prime Day," *Mobile Marketer*, July 1, 2019.

as a response to consumer fatigue with ads and the desire for a predictable revenue stream for companies.

Apple Music is a prime example of a pure subscription model in mobile streaming. Launched in 2015, it quickly grew to tens of millions of subscribers, reaching around 88 to 93 million subscribers by 2023,²³ making it the second largest music streaming service after Spotify. Apple Music does not offer a free tier aside from time-limited trials. In return, subscribers get an ad-free experience and a comprehensive catalog of music. Apple Music deliberately ties into Apple's broader strategy: The service reinforces the value of Apple's hardware ecosystem (phones, earbuds, operating system, etc.), and it generates recurring revenue that complements hardware sales. Even in subscription services, mobile personalization plays a key role. This is similar to Netflix's tailored recommendations or Spotify's custom playlists that add value for paying users by catering to individual tastes. Apple can afford not to chase advertising dollars here because its primary business is selling devices and services as a package. Indeed, Apple has become known for using one arm of the business to bolster another. By providing appealing services like music (or shows via Apple TV+), Apple adds value to its iPhones and retains customers within its closed ecosystem, even if these services largely break-even or even operate at a loss. It then profits when those customers buy the next device or additional paid services. This is a strategic use of subscription content to support hardware sales, a different approach from companies that rely solely on content for profit. In an age of platform mobility, Apple very well may be the last company to be able to offer a model such as this. The legacy and strong branding/marketing campaigns of Apple have cultivated a very niche and devoted audience.

YouTube has blended subscription with its primarily ad model by introducing YouTube Premium, formally known as YouTube Red. For a monthly fee, YouTube Premium offers an ad-free viewing experience, background play for videos on mobile, allowing continued audio while using other apps, and access to YouTube Music Premium. While the majority of YouTube's billions of users remain on the free, ad-supported tier, the premium option caters to a segment of users who are willing to pay for convenience and no ads. This dual approach lets YouTube monetize both groups: Ad-tolerant users generate advertising revenue, and ad-averse power-users generate subscription revenue. Similarly, Twitch employs Twitch Turbo for an ad-free experience and encourages viewers to subscribe to individual channels (more on that

²³ Joseph D'Souza, "Apple Music Statistics and Facts (2025)," *Electro IQ*, March 17, 2025.

in the next section), thereby mixing ad and subscription income. The move to incorporate subscriptions reflects an understanding that not all revenue should be left on the table for the sake of scale; some users can be “upsold” to higher-margin direct payments. It’s a diversification that can make the platform less dependent on fluctuations in ad markets.

From the user viewpoint, subscriptions can be both enabling and limiting. On one hand, they free the user from advertising and often provide higher-quality service, such as higher audio quality on music services and exclusive content on some video platforms. On the other hand, too many subscriptions can lead to subscription fatigue. As more platforms partition content behind paywalls, users must decide which ones are worth paying for. In the realm of streaming video, we see an analogous fragmentation with Netflix, Disney+, Amazon Prime Video, etc., though in mobile-centric platforms the fragmentation is more along functional lines. In other words, music versus video versus gaming content. This can potentially drive some users towards unofficial or pirated content if the paywalls become too high, but in practice many have shown willingness to pay for convenience and high quality, as evidenced by the growth of services like Netflix or Spotify.

For the platforms, a steady subscription base is appealing because it provides predictable revenue and reduces reliance on the volatile advertising market, which can fluctuate with economic conditions. However, to attract and retain subscribers, platforms must consistently offer value through original content, better features, or an ecosystem of benefits. Amazon is a prime example of this, as they bundle Prime Video with free shipping and other perks under a single Amazon Prime subscription. This creates a pressure to continually invest in content and features, which is why we see, for example, YouTube investing in original shows to entice Premium subscribers, or Apple bundling Apple Music in discounted bundles with other Apple services like Apple One.

Subscription models may seem more straightforward than advertising, but they still involve power dynamics and capitalizing on consumer behavior. Companies will price discriminate by offering student or family plans to maximize sign-ups and they will engage in lock-in strategies, such as playlists or content libraries that are hard to transfer if one leaves a service. And even if they are not selling ads, they will still collect data on subscribers to inform future offerings. Moreover, subscription services can concentrate cultural power. For instance, if certain music is only on Apple Music due to exclusive deals, that gives Apple gatekeeping power over that content.

Generally, though, subscription in mobile streaming coexists with free options, forming a tiered ecosystem of monetization. This model is widespread across streaming sectors: Video platforms like Netflix or Disney+ thrive on subscriptions, while music services like Spotify blend free and premium tiers, and even social media (e.g., YouTube Premium) uses subscriptions to upsell an enhanced experience.

Microtransactions and the Creator Economy

A third monetization strategy revolves around microtransactions and virtual gifting, central to the creator economy. Mobile streaming platforms, especially those centered around live content like Twitch, TikTok, YouTube Live, and Instagram Live, have embraced microtransactions and virtual gifting as core monetization strategies. These systems are central to what's now called the creator economy, where individuals earn income from their content, though often on terms heavily dictated by the platform.

For example, in addition to allowing users to add links to advertised products and services in their bios and posts, TikTok has also added the ability for audiences to purchase products featured in videos directly via the video.²⁴ Virtual gifts and in-app currencies let viewers spend real money to purchase coins, badges, or animated icons to send during live streams. TikTok coins, YouTube's Super Chat, Instagram's badges, and the aforementioned Bits on Twitch are all examples. These interactions serve dual purposes. For the creator, they reward and boost visibility or recognition for the viewer for donating to the creator. From the platform's perspective, this is quite lucrative as it monetizes user-to-user engagement while taking a sizable cut of each transaction. Importantly, this setup relies on parasocial relationships and fans paying for attention, validation, or affiliation. Crucially, the highly personalized nature of users' content feeds, particularly the constant surfacing of favorite creators, helps cultivate the kind of loyal engagement that fuels these microtransactions. This is another facet of mobile personalization as each user's feed is tuned to strengthen the parasocial bond that platforms then monetize.

Subscriptions to creators further expand the model. Twitch pioneered this approach, letting viewers subscribe monthly for perks like custom emojis and subscriber-only chat. YouTube and TikTok have since followed with their own versions. These subs are marketed as direct support for creators,

²⁴ Ivan Mehta, "TikTok Shop Officially Launches in the US," *TechCrunch*, September 12, 2023.

but revenue splits (often 50/50) and policy changes make it precarious. Even top streamers face sudden changes, like Twitch's decision to roll back 70/30 splits for high earners, citing infrastructure costs and increased ad revenue as compensation.

Social commerce and affiliate integration are also on the rise. Platforms like TikTok and Instagram now let creators tag products in content, turning livestreams or short videos into shoppable ads. The platform typically takes a cut, reinforcing a model where content, commerce, and monetization blur together. This strategy, sometimes called *shoppertainment*, is especially prevalent in live shopping streams.

These forms of monetization signal a shift. Instead of only selling audience attention to advertisers, platforms now monetize individual fan engagement and creator labor directly. Virtual currencies play a key role here. They introduce a layer of abstraction—users convert money into coins, which can't usually be withdrawn—encouraging spending and locking value into the ecosystem. In some ways, this resembles scrip systems which harken back to coal mine towns in the early eighteenth and nineteenth centuries, reinforcing platform dependency. Though framed as empowering for creators, these systems carry deep inequalities. Small numbers of creators earn a living; most do not. The labor of streaming, posting, and cultivating an audience generates significant platform value, but creators remain subject to algorithm shifts, policy changes, and commission structures which they do not control. This situation demonstrates a clear power imbalance. While individual creators are dependent on the platform's algorithms and policies, the platform wields structural power over who can monetize and how. The result is a precarious labor economy where platforms extract value from both sides, content creation and audience engagement, while retaining power over the entire system.

Monetized Cross-Subsidies

A fourth layer to monetization is how mobile streaming platforms often are part of larger corporate ecosystems, and how cross-subsidy strategies are used. We've touched on this with Apple (using services to bolster device sales) and can see it with others:

Amazon acquired Twitch in 2014 and integrated it into its Prime membership. Prime subscribers get one free Twitch channel subscription per month, which Amazon pays the streamer as if it were a normal subscription. This effectively means Amazon is subsidizing Twitch streamers to add value to

Prime, which is mainly justified by other revenue. Similarly, Amazon might accept lower margins on Twitch because Twitch runs on AWS, potentially driving business to AWS or benefiting from internal cost offsets. It's an example of using one platform's strength to support another. In other words, Amazon's retail profits and cloud profits can subsidize Twitch's growth.

Google distributes Android freely (open-source) and many of its services for free to ensure its presence on mobile—a strategy explicitly designed to undercut Apple and make Google the gatekeeper on mobile. In return, Google reaps massive advertising revenue from the mobile usage of YouTube, search results, and Android usage, for example. It's a cross-subsidy where a free mobile OS leads to more mobile internet usage, which leads to more searches and video views that Google can monetize. In a similar vein, YouTube itself is sometimes seen as subsidized by Google's deep pockets. For years YouTube operated at a financial loss or slim margin due to infrastructure costs and efforts to grow, but Google's other revenue streams allowed it to keep investing until YouTube became profitable. The free or inexpensive parts lure users in; the monetized parts extract the value.

These cases show that “streaming” isn't a standalone industry at all, but an infrastructural logic entwined with other industries (e.g., e-commerce, hardware, cloud services) as tech giants leverage one arm of their business to boost another.

In some arenas, particularly music streaming, content is seen as a loss leader. In the music streaming space, companies like Amazon and Apple use music as part of a larger strategy—i.e., to boost sales of Echo speakers and iPhones or encourage Prime subscriptions. They might have lower margins on music rights compared to a standalone company like Spotify, but they don't mind as much because the goal is an ecosystem lock-in. For Apple, someone on Apple Music might be less likely to switch to Android because their playlists and familiarity are in Apple's walled garden. For Amazon, offering Prime Music is a value-add to keep Prime subscriptions high. These indirect monetization effects are harder to quantify but are crucial in understanding why tech giants all want a piece of streaming media. It's not only about direct revenue; it's also about strategic control of user engagement, which can be turned into various forms of value.

By employing cross-subsidies, big platforms can outcompete smaller ones that do not have alternate revenue streams. This raises barriers to entry. For instance, a startup that makes a streaming app might struggle to monetize if users now expect free access because incumbents offer content

free, perhaps even despite knowing they monetize elsewhere. Once drawn in, the platform can monetize them in the domain where it has strength. This is why we often see tech giants entering each other's territories. Google in music and video, Amazon in video, Facebook in e-commerce, etc., each may offer something at low cost because they have another way to profit after capturing the user. This can lead to predatory pricing or unfair competition debates. For example, Spotify has argued that Apple's dual role as platform owner and competitor service is unfair, since Apple can afford to take a loss on music streaming which Spotify cannot.²⁵

In the contemporary mobile streaming ecosystem, monetization is multifaceted and constantly evolving. By the time you read this book, there is a high likelihood that these models have transformed in some way or new models have been introduced. Platforms rarely rely on a single source of revenue; rather, they build hybrid models to maximize profit: Ad-supported free usage casts the widest net for users and monetizes through selling attention. Premium subscriptions convert a fraction of users into direct payers for enhanced experience. Microtransactions and virtual economies tap into social and community dynamics to generate revenue from user interactions. Data-driven strategies improve all of the above and open potential for indirect monetization. Integration with commerce turns platforms into marketplaces (e.g., Instagram Shopping, TikTok integrating retail), further diversifying income. Ecosystem leverage and subsidies ensure that even if one service isn't highly profitable, it serves a strategic role in a larger profitability picture.

For users, this means their experience is often segmented. One can choose to "pay with attention" (watch ads) or pay with money for convenience; one can use the service freely but have one's data heavily mined or pay a premium that perhaps involves slightly less data exploitation (though the protections provided to such subscribers are incomplete at best). For creators, multiple avenues exist to earn (ad revenue shares, fan subscriptions, sponsorship deals facilitated by their fame on the platform), but all paths keep creators dependent on the platform's rules and revenue sharing frameworks.

From a critical perspective, one might note that these monetization methods rely on commodifying not just content but also audiences, data, and social interactions. The user base is segmented and sold to advertisers; user data is packaged into marketable insights; even the act of connecting with or influencing others becomes a revenue-generating service. In effect, mobile streaming platforms continually create new commodity forms out

²⁵ "The Case," *Time to Play Fair* (website), Spotify, accessed May 12, 2025.

of digital life. Whether through selling access to attention, charging for enhanced experiences, or skimming off user-to-user payments, platforms relentlessly seek to extract value from every moment of engagement. All of these strategies underscore a fundamental imperative of platform capitalism: *to leave no user activity unmonetized.*

APIs and Platform Capture

Lastly, beyond user-facing revenue models, platforms monetize through controlling technical integrations. This is what we call APIs and platform capture. Mobile streaming platforms do not exist in isolation. They function within broader platform ecosystems and often interconnect (or deliberately don't interconnect) with other services. The degree of integration versus isolation is a strategic choice that significantly shapes user experience and monetization opportunities. APIs, or application programming interfaces, are a key technical mechanism in this context, as they determine how easily different services can communicate and build upon each other. A deeper look at APIs and their integrations reveal how platforms balance openness, to encourage growth and use of their platform, with closure, to maintain control and exclusive benefits. It also highlights how big platforms leverage their ecosystems to fortify their market position, creating ecosystems that lock users in. While the previous sections focused on direct revenue streams, this section examines how controlling the very connections between services (via APIs) translates into monetization and market power.

Many mobile streaming platforms make use of integration to broaden their reach. YouTube, for example, provides embedding APIs that allow its videos to be played on countless websites and apps outside YouTube itself. This openness was instrumental in YouTube's early dominance. Users could encounter YouTube content on blogs, news sites, social media feeds, etc., with YouTube reaping the benefits of increased views and data collection and ads, if served. Similarly, Spotify and Apple Music have APIs that let other apps fetch song information or integrate music playback. For instance, Instagram allows linking a song snippet to a Story via Spotify. These integrations enhance user convenience and propagate the platform's presence across digital spaces. They also generate real or imagined network effects: If your favorite fitness app integrates with Spotify for workout playlists, you are more likely to subscribe to Spotify, for instance.

However, with greater maturity, platforms often turn toward closure and exclusivity to cement their power. Many contemporary strategies involve

making the platform a one-stop-shop so that users have less reason to use anything else. Facebook's integration of multiple functions (messaging, video, shopping) into one app family is a case in point. In the mobile streaming realm, we see analogous moves. Instagram and Facebook heavily promote their own Reels feature to compete with TikTok, even allowing cross-posting of Reels between the two Meta-owned networks, but they do not make it easy to share that content to outside platforms. TikTok, while it allows sharing of video links, prefers to keep users inside its app where the "For You" feed can continue to serve videos. It is important to note that if you are to click a TikTok link without the app, it often prompts you to download the app. Furthermore, platforms can impose technical or policy barriers. Twitter historically provided APIs for third-party clients but later restricted them to force users back to the official app where ads could be served; similarly, Facebook cut off some API access (like friend graphs) to outsiders, citing privacy but also arguably to prevent competition.

APIs shape monetization in that they define who can innovate on top of the platform and where value flows. An open API can stimulate an ecosystem of complementary apps. For example, Twitch's API enables a whole market of chatbot tools, stream overlays, and extensions that enrich the Twitch experience, making Twitch more attractive to streamers and viewers, which indirectly boosts monetization for Twitch. But open APIs can also create vulnerabilities. If third-party apps can show content without ads or capture user attention elsewhere, the platform might lose revenue. This could be pointed to as one of the reasons why Twitch, for example, has stopped counting third-party integrations toward viewers in any given stream. As such, this could be why many platforms gradually tighten API accessibility as they scale. We can interpret this as platforms wanting to internalize all interactions so that any profit potential remains in-house. The rise of closed-API apps in mobile is a broad trend. In practice, this means that each major company tries to channel user activity into its own app, curtailing interoperability that might let a rival benefit.

Cross-platform integration also involves hardware and software coordination. A salient example is how certain services are tightly integrated with specific devices or operating systems. Apple leverages this with features like Handoff/Continuity that make it seamless to move from watching or listening on an iPhone to doing so on a Mac or iPad—but only for Apple's own services or approved partners. By making the experience smoother within an ecosystem than outside it, companies incentivize users to stay within

their suite of products. Another example is exclusive compatibility. Apple's AirPlay casting feature works more readily with Apple's own streaming services, while Google's Chromecast was originally optimized for YouTube and Netflix, and Google limited Amazon Prime Video's casting until business disputes were resolved.²⁶ These technical choices have economic motives, as they subtly push users towards certain content sources over others.

We also see strategic non-integration as a form of competition. A case in point is how Amazon and Google feuded for years, resulting in YouTube being unavailable on Amazon's Fire TV devices for a time, or Amazon refusing to sell Google's Chromecast on its store. Effectively, each company was trying to deny the other access to its user base in order to bolster its own platform's edge. Likewise, Tencent's and Alibaba's ecosystems in China historically blocked each other's links to keep users from migrating out (e.g., WeChat blocking Douyin [TikTok] links). These battles demonstrate that integration or its refusal can be a tool of power: Controlling points of connection is as important as controlling content.

For users, a high degree of integration can be a double-edged sword. Integration brings convenience: the ability to log into many apps with one account (your Google or Facebook login), to share content easily between services, or to have a unified experience across devices. But it can also lead to diminished choice and privacy. If one company's ecosystem handles everything, that company wields tremendous influence over the user's media intake and captures a comprehensive data profile. When all your social contacts, entertainment, and communication occupy one platform, the cost of leaving that platform (the switching cost) becomes very high. This lock-in is exactly what companies aim for, as it secures their user base against competitors. Indeed, a further downside of this model is that it can be anti-competitive and stifle innovation; new entrants might be shut out because the dominant platforms either don't allow interoperability or can copy and integrate any new feature as Instagram did in copying Snapchat's Stories, or YouTube in cloning TikTok's short-form vertical video format with "Shorts." In essence, large platforms use their integration capabilities to create a moat around their user base.

The role of APIs in monetization is also evident in how third-party developers are allowed to participate in the platform's economy. Platforms like Facebook and YouTube share some ad revenue or allow third parties to develop for the platform (as with Facebook's Instant Games or YouTube's

²⁶ "Amazon and Google Deliver on TV Peace Deal," *BBC News*, July 10, 2019.

partner program), but they keep a significant portion of revenue and full access to data. If a platform instead kept its API entirely closed, it might slow growth but ensure any monetizable extension of the platform is built internally (thus capturing all value). We can see mixed models: Apple's App Store is a kind of API that lets developers create apps for the iPhone—Apple takes a 30 percent cut of any digital sales and has final say on what apps appear. This has been immensely profitable for Apple and extends the iPhone's functionality, but Apple carefully polices this ecosystem to prevent apps that would substitute its own services—for example, restrictions on game streaming apps, as mentioned, or enforcing in-app purchase rules. More insidiously, Blizzard's World of Warcraft is known for taking popular third-party add-ons and integrating popular features into the game itself with no recognition or compensation to the original developer.

In summary, cross-platform integration and APIs are not merely technical matters; they are planned and intentional business decisions. They determine who gets to build on the platform and who reaps the benefits. Open integrations can drive growth and enrich user experience, but too much openness can undercut a platform's monetization or allow competitors in; thus, most platforms adopt a strategy of selective integration. They open up enough to become pervasive but not so much that they lose control of user data or revenue flows. As platforms mature, there is a clear trend toward asserting more control through strategies such as tightening API use, cloning external innovations in-house, and promoting closed-loop usage. The endgame for each major player is to have users living their digital lives within that platform's ecosystem as completely as possible—which is the ultimate form of monetization and power, as it means the platform mediates a large share of social and media experience and can channel those experiences into profit.

Conclusion

Mobile streaming has swiftly transformed the media landscape and woven itself into the daily lives of billions, enabling an always-on, personalized stream of content that flows across devices and borders. This mobile personalization of media, where content is tailored to users in real time, is not just a user-friendly feature but the very engine of monetization that we explored in this chapter. By examining this phenomenon, we discern that its story is not just about technology or entertainment; it is fundamentally about power, profit, and control in the digital age. Platforms like TikTok,

Twitch, Instagram Reels, YouTube, Apple Music, and others exemplify the new breed of media companies whose operations hinge on data-driven, networked capitalism. They have embraced platform mobility—making content and data accessible everywhere—not simply to please users but to maximize engagement and revenue. They have turned user activities and social interactions into commodities, capturing attention and data and converting them into advertising placements, subscription fees, or transaction cuts. In doing so, they blur the line between leisure and labor, consumption and production, as users become unwitting workers in the generation of the very data that the platforms monetize.

Throughout this chapter, we have seen that monetization strategies are at the heart of why mobile streaming looks and feels the way it does. The endless feeds, the algorithmic recommendations, the push notifications are all optimized to serve the business model; typically advertising or engagement-based sales. Even as new models emerge, the pull of commercial imperatives remains. This raises the question: *To what extent should our daily communication and entertainment spaces be governed by the pursuit of profit?* When virtually every form of expression, from a TikTok dance video to a heartfelt personal vlog, takes place on corporate platforms, those corporations set the rules of visibility, acceptable speech, and revenue sharing. This chapter has not focused on free speech or content moderation issues, but it is implicit that the power to monetize is also the power to moderate. Platforms, undoubtedly, will allow or disallow content based in part on what is good for business. Avoiding scandal, pleasing advertisers, and complying with laws to avoid shutdown are always part of the conversation, if not the conversation itself. Thus, mobile streaming is intertwined with questions of democracy and autonomy in the digital public sphere.

The story of contemporary mobile streaming is a story of pervasive connectivity tangled with commercial agendas. It is a story of how our most casual daily habits—checking an app during a commute, sharing a video clip with friends, listening to a favorite song on the phone—are embedded in a global network of economic transactions and power relations. Throughout this chapter, we hope to have peeled back the screen to reveal the currents of capital and control that flow beneath, shaping the infrastructure, content, and uses of mobile streaming. This understanding is crucial for scholars, policymakers, and citizens alike, as we grapple with ensuring that the networked digital media, which now mediates so much of our social life, serves the public good and not only corporate interests. Mobile streaming has

unleashed incredible creativity and convenience, but it has also entrenched new forms of corporate power. This isn't one singular industry's story but a transformation cutting across entertainment, communication, and tech industries via a unified platform logic. Negotiating that duality, harnessing the benefits of ubiquitous connectivity while reining in its exploitative excesses, will be a defining challenge for years to come. Ultimately, recognizing mobile personalization and monetization as instruments of infrastructural power is crucial if we are to challenge and balance the influence these platforms wield moving forward.